K-FIELD DISCOVERED PATHWAYS

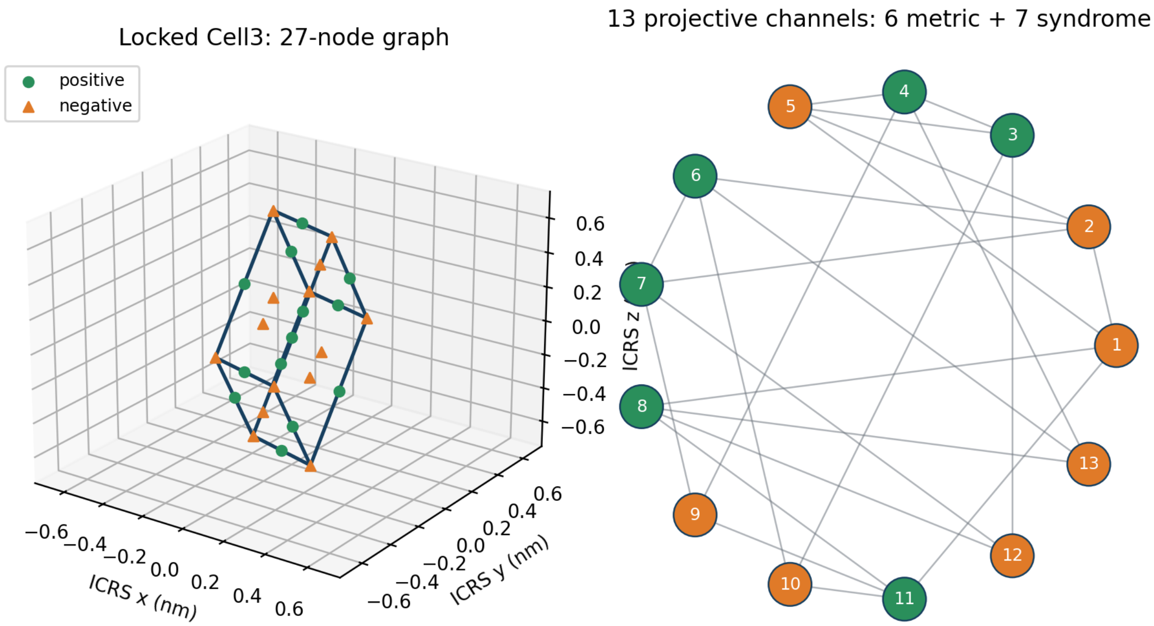
An exhaustive multidimensional graph treatment of Cell3 geometry, finite projective incidence, Devon polarity, overdetermined metric decoding, Mass-Bridge reciprocal invariants, Lucas closure, recursive spectra, and serialized nonlocal routing

Principal Investigator: P. Dwayne Esterline

BS, Huntington University (1999) | MBA, Indiana Wesleyan University (2008)

Krytronx, LLC

K-Field Discovered Pathways: Multidimensional Graph Architecture



Title figure. The left panel is generated from the locked physical Cell3 basis and displays all 27 local graph nodes in ICRS nanometer coordinates. The right panel displays the 13 antipodal projective channels derived from the same ternary address set. No schematic or arbitrary coordinates are used.

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Abstract

This report treats the complete K-Field Cell3, Mass-Bridge, Lucas Supergroup, Devon Super Sheet, recursive spectrum, and serialized address corpus as one exhaustive multiplex graph problem. The same finite state set is read in several mathematically compatible ways: as a 27-node open Cartesian graph; as the affine space AG(3,3); as 13 antipodal directions in the projective plane PG(2,3); as an overdetermined 13-channel measurement code for a six-parameter symmetric metric; as a 729-volume Lucas closure with 55 laminar sheets; and as a recursively scaled hierarchy whose volume, face, edge, and point channels grow by 729, 81, 9, and 1 per closure step. These views are not separate analogies. They are different graph projections of the same locked relational object.

The central discovered relationship is that Devon checkerboard polarity organizes the projective measurement code exactly. The six positive projective classes are the six face-diagonal directions and form an invertible six-channel metric decoder. The seven negative classes are the three coordinate axes plus four body-diagonal classes and form the complete overdetermination syndrome. Once the metric is recovered, the Mass-Bridge ratios follow as reciprocal graph invariants. At larger scale, the number of Devon Super Sheets equals the nullity of the corresponding open cubic Emily adjacency operator at every exact Lucas closure. The recursive morphology then becomes a dimension-graded renormalization system, while scalar serialization supplies identity-preserving but intentionally nonlocal routing.

The report develops these results from the numerical kernel, proves the exact identities, supplies dimensionally correct figures and complete node and plane ledgers, and defines experimental and computational pathways that follow from the combined graph architecture.

1. Reader Orientation: One Object, Many Graphs

A graph is a set of vertices together with declared relationships called edges. A multidimensional graph problem allows several edge types to coexist on the same vertices. In the present system, geometric adjacency, antipodal pairing, projective incidence, laminar membership, containment, reciprocal coupling, recursive inheritance, and scalar ordering all define different edges or hyperedges. Treating them together is essential because a relationship hidden in one projection can become exact and simple in another.

```
G_multi = (V, E_geometric, E_antipodal, E_projective, E_sheet, E_containment, E_spectral,
E_recursive, E_rho)
```

The vertex sets occur at several resolutions. The local Cell3 has 27 ternary nodes. The Lucas Supergroup has 729 Cell3 volumes, 1,000 merged unrefined corner nodes, 6,859 refined half-step states, and 55 constant-lambda sheets. The projective quotient has 13 unoriented nonzero directions. Recursive closure produces further levels without changing the underlying grammar. The objective is not merely to count these objects, but to determine which transformations preserve information, which subgraphs carry physical parameters, and which residual spaces expose error or new dynamics.

Foundational principle. A relationship is treated as strongest when it is exact under independent descriptions. The work therefore repeatedly compares direct geometry, reciprocal geometry, finite incidence, graph topology, polarity, spectral nullity, and recursive scaling. Exact agreement across these layers is the principal discovery criterion.

2. Locked Cell3 Geometry

2.1 Direct basis, metric, and reciprocal metric

The locked physical Cell3 is an oblique parallelepiped in ICRS Cartesian coordinates. Its three basis vectors are the columns of B. A primitive address vector u is converted to a physical displacement by B u. The direct metric G=B^T B contains all squared lengths and mutual dot products. The reciprocal metric R=G^-1 measures the corresponding reciprocal quadratic form.

```
B = [[0.3275012420558906, 0.0353307758438284, -0.01012931762672882],
[-0.05489700788400644, 0.2073198504401918, -0.3897877023662599],
```

```
[-0.03748750289472991, -0.5474010853885081, -0.6897344083672676]] nm
G = B^T B; R = G^-1
```

Matrix	row 1	row 2	row 3
G (nm^2)	0.11167605789605	0.0207103332806058	0.0439372350962198
	0.0207103332806058	0.343877732392787	0.296392758961523
	0.0439372350962198	0.296392758961523	0.627770610077296
R (nm^-2)	9.20802552945755	0.00153480242819255	-0.645188035699939
	0.00153480242819259	4.90339831236799	-2.31517558441639
	-0.645188035699939	-2.31517558441639	2.73117127471169

Quantity	a channel	b channel	c channel
edge length (nm)	0.334179679059	0.586410890411	0.792319765043
opposite-face spacing (nm)	0.329546529695	0.451597379721	0.605097741519
included angle (deg)	ab = 83.933483012313	ac = 80.448130018338	bc = 50.363231813756
primitive face area (nm^2)	bc = 0.357809539129	ac = 0.261106235790	ab = 0.194869165460
Cell3 volume (nm^3)	0.117914891911745		

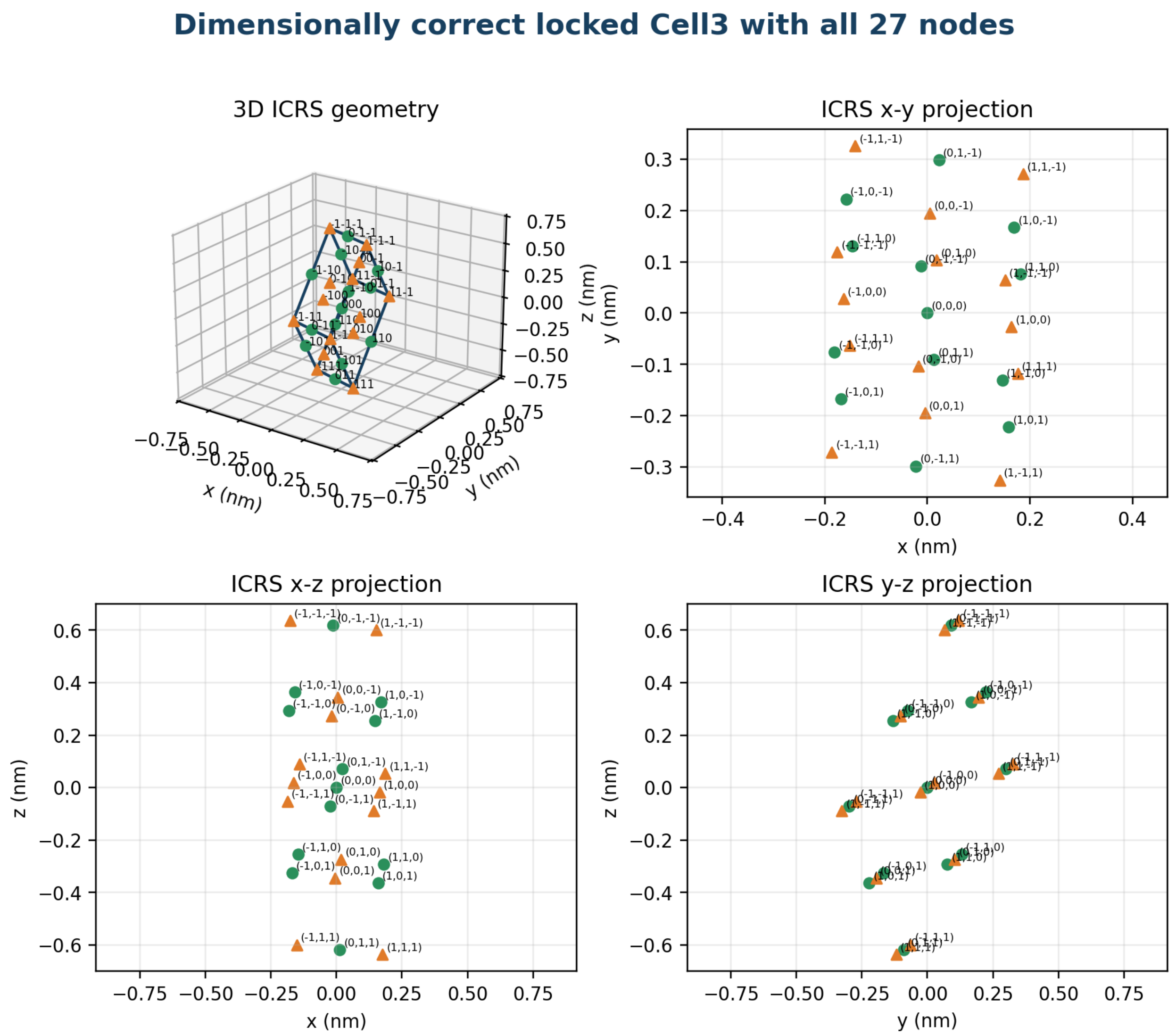

Figure 2.1. Dimensionally correct locked Cell3. The 3D view and three orthographic projections show every address u in {-1,0,1}^3 at the physical position r(u)=B u/2. Green circles are positive even-lambda nodes; orange triangles are negative odd-lambda nodes. Every node is labeled by its complete ternary address.

2.2 The complete 27-node scaffold

The local node set consists of all ternary address triples. The position, laminar grade, and polarity are

```
u in {-1,0,1}^3; r(u)=B u/2; lambda(u)=u1+u2+u3; chi(u)=(-1)^lambda
```

The number of zero coordinates classifies the physical location. Zero zero-coordinates gives eight corners; one zero gives twelve edge centers; two zeros gives six face centers; three zeros gives the body center. Antipodal inversion u -> -u pairs every noncentral node. Because lambda changes sign but preserves parity, antipodal partners have the same checkerboard polarity.

Node class	zero coordinates	count	projective antipodal classes	polarity structure
corner	0	8	4	negative
edge center	1	12	6	positive
face center	2	6	3	negative
body center	3	1	1	positive center

2.3 Six exact boundary planes

Each primitive coordinate has a negative and positive boundary plane. In physical coordinates the plane equations can be written with the inverse basis:

```
e_i^T B^-1 r = +/- 1/2
```

Each plane contains exactly nine of the 27 nodes. The plane normal is a reciprocal-basis direction. The face area and opposite-face spacing are linked by volume = area x spacing. These facts make the plane system simultaneously geometric, reciprocal, and graph-theoretic: fixing one address coefficient selects a 3 x 3 induced subgraph.

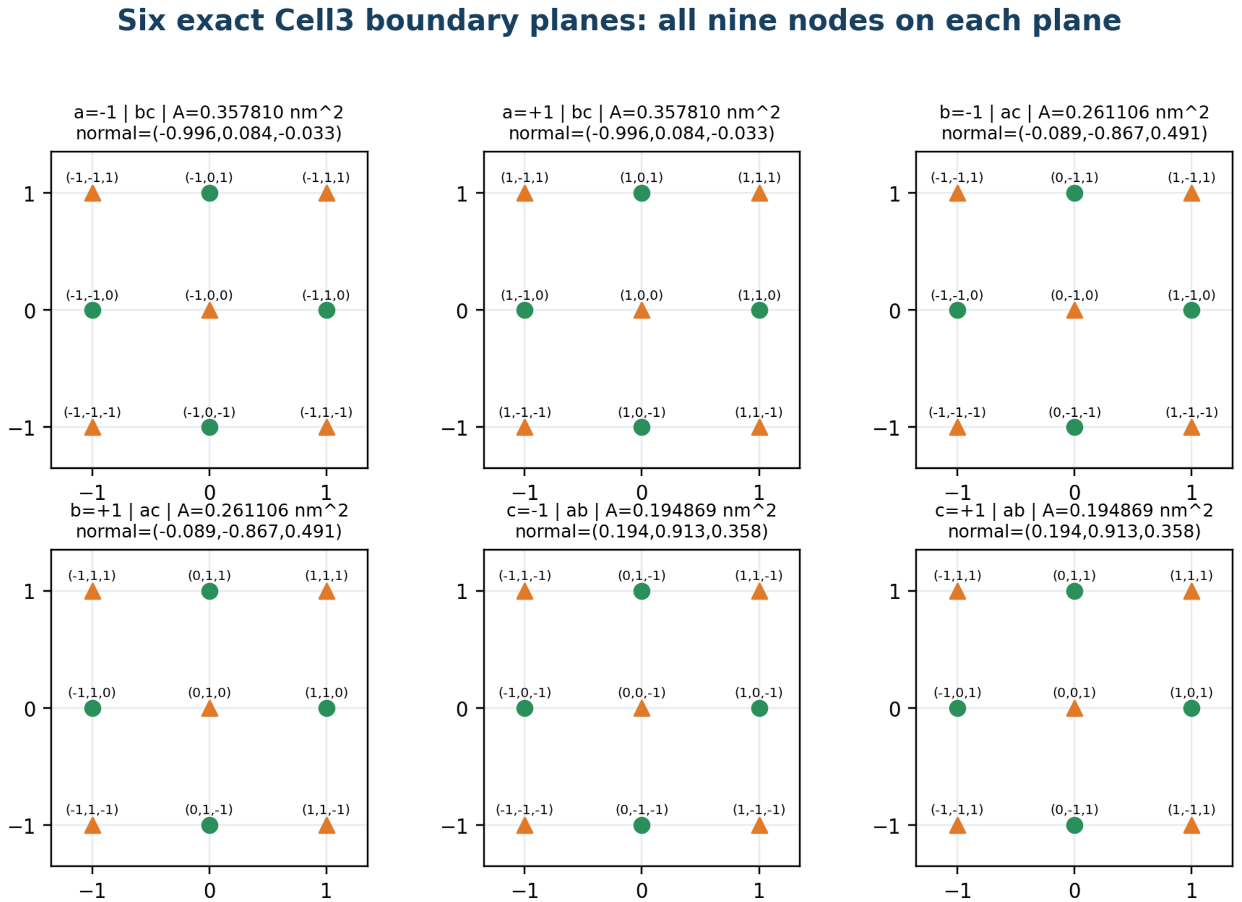

Figure 2.2. Complete six-plane atlas. Every panel is generated from the exact plane incidence list. All nine addresses are shown, together with face family, physical area, and ICRS unit normal.

3. Cell3 as an Open Cartesian Graph

3.1 P3 x P3 x P3

Geometric nearest-neighbor adjacency changes exactly one address coefficient by one step. The local graph is therefore the Cartesian product of three three-vertex path graphs, P3 x P3 x P3. It contains 27 vertices and 54 edges. Because it is connected, its graph cycle rank is E-V+1=28.

```
V=27; E=3(3-1)3^2=54; beta1=54-27+1=28
```

Every edge changes lambda by one and therefore flips polarity. The seven laminar populations are 1,3,6,7,6,3,1. The counts of edges crossing successive sheet boundaries are 3,9,15,15,9,3. The graph is thus a layered bipartite current carrier: all geometric motion alternates between positive and negative states.

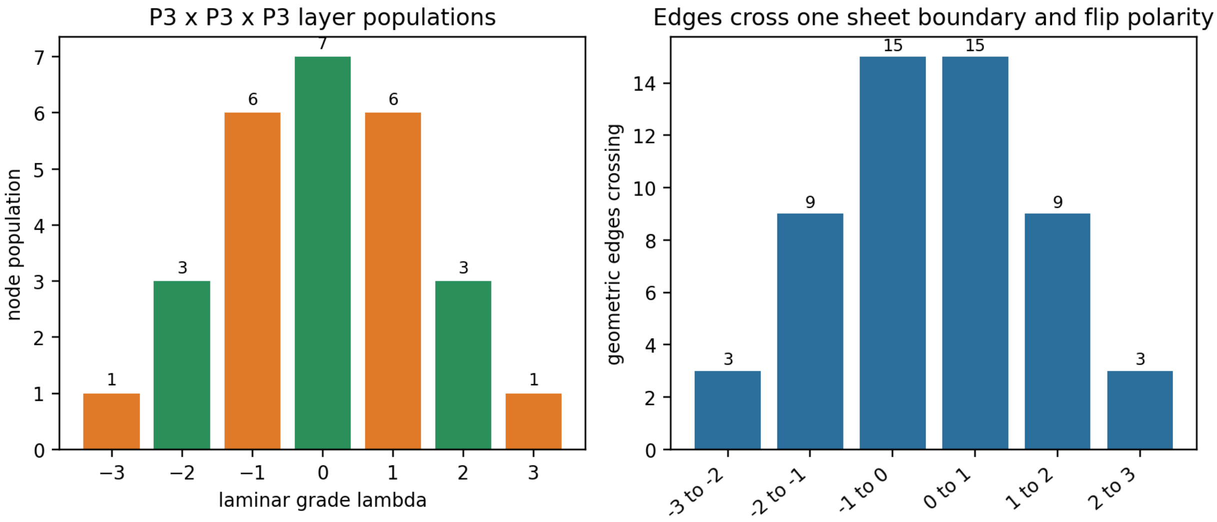

Figure 3.1. Exact laminar populations and inter-sheet edge counts for the open Cell3 graph. The profiles are generated by exhaustive enumeration of all 27 nodes and 54 nearest-neighbor edges.

3.2 The same state set as AG(3,3)

The symbols -1,0,1 can also be read as the three elements of the finite field F3. Under addition modulo three, the 27 addresses form the three-dimensional affine space AG(3,3)=F3^3. This periodic interpretation adds affine lines and planes that are not physical nearest-neighbor edges, but it preserves every address exactly. The open graph and finite affine geometry therefore coexist on the same vertex set.

This dual reading is a primary source of hidden relationships. The open graph records local physical adjacency and boundary. The affine geometry records periodic incidence, directions, and quotient structure. Neither replaces the other; the multiplex system retains both.

4. The 13-Channel Projective Plane

4.1 Antipodal quotient

Remove the zero address and identify every nonzero vector with its negative. The 26 directed nonzero vectors become 13 unoriented projective directions:

```
(F3^3 - {0})/{+/-1} = PG(2,3); (3^3-1)/(3-1)=13
```

These 13 directions are exactly the projective representatives stored in the serialized address system. The number 13 is therefore forced by the complete local address geometry. It is not selected externally and it does not omit any nonzero local direction.

4.2 Polarity incidence matrix

For projective representatives u_i and u_j, define an incidence when their ordinary dot product is zero modulo three. The resulting 13 x 13 symmetric matrix N is the polarity incidence matrix of the projective plane of order three. Every row and column contains four incidences. Exhaustive multiplication gives

```
N^2 = 3 I + J
```

where J is the all-ones matrix. The spectrum is 4 once, +sqrt(3) six times, and -sqrt(3) six times. The supplied projective transform is M=3N-J, so

```
M^2 = 27 I - 2 J
```

and its spectrum is -1 once, +3sqrt(3) six times, and -3sqrt(3) six times. These eigenvalue multiplicities reproduce the exact 1+6+6 hidden projective decomposition.

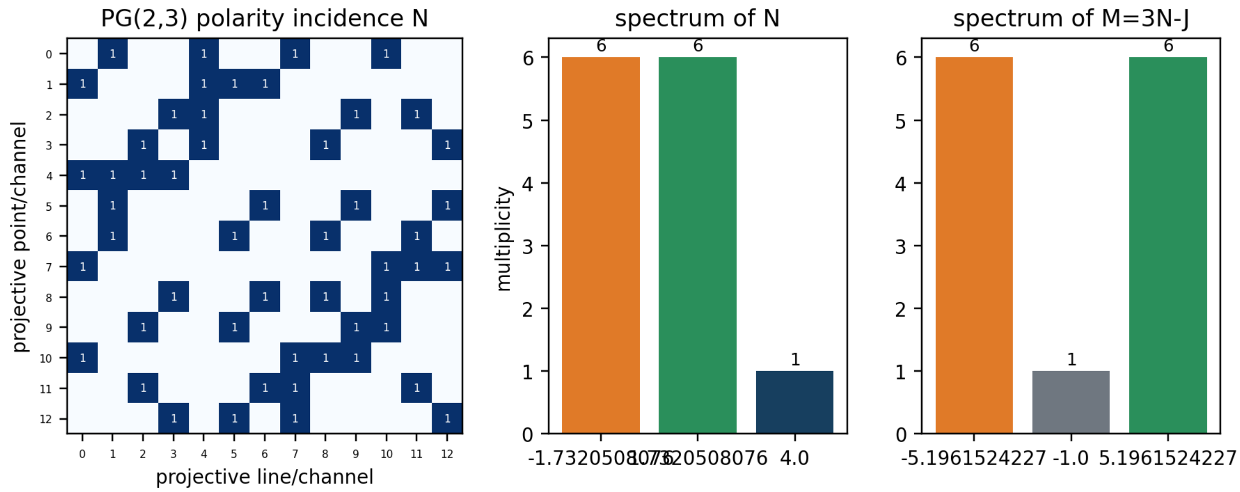

Figure 4.1. Exact projective incidence and spectra. The incidence matrix is generated directly from the 13 supplied representatives. The spectra are numerical evaluations of the exact identities N^2=3I+J and M=3N-J.

4.3 Levi graph and the 13 -> 27 -> 55 sequence

The point-line Levi graph of PG(2,3) has 13 point vertices, 13 line vertices, and 52 incidence edges. Its cycle rank is 52-26+1=27, which equals the number of affine states in AG(3,3). Separately, the periodic coordinate graph C3 x C3 x C3 has 27 vertices and 81 edges, so its cycle rank is 81-27+1=55. That equals the number of refined laminar sheets in one Lucas Supergroup.

```
13 projective directions -> 27 affine states -> 55 periodic cycles/sheets
```

This sequence links quotient direction, complete local state, and periodic cycle space. It is not a numerical coincidence assembled from unrelated counts: each number is generated from the same F3^3 address geometry under a different graph operation.

Object	vertices	edges	cycle rank	cross-layer identity
PG(2,3) Levi graph	26	52	27	equals |AG(3,3)|
periodic C3 x C3 x C3	27	81	55	equals Lucas refined sheet count
open P3 x P3 x P3	27	54	28	local physical cycle rank

5. Devon Polarity as the Metric-Syndrome Partition

5.1 Exact 6+7 split

Evaluate chi(u)=(-1)^(u1+u2+u3) on the 13 projective representatives. Six classes are positive and seven are negative. The positive classes are exactly the face diagonals (0,1,+/-1), (1,0,+/-1), and (1,+/-1,0). The negative classes are the three coordinate axes and the four body-diagonal classes.

This is the antipodal reduction of the complete local Cell3 classification. The 12 positive edge centers become six positive projective classes. The six negative face centers and eight negative corners become seven negative classes. Including the positive body center leaves 13 positive and 14 negative local nodes, so the net local polarity is -1. That same residual persists through every exact Lucas closure in the Devon Super Sheet system.

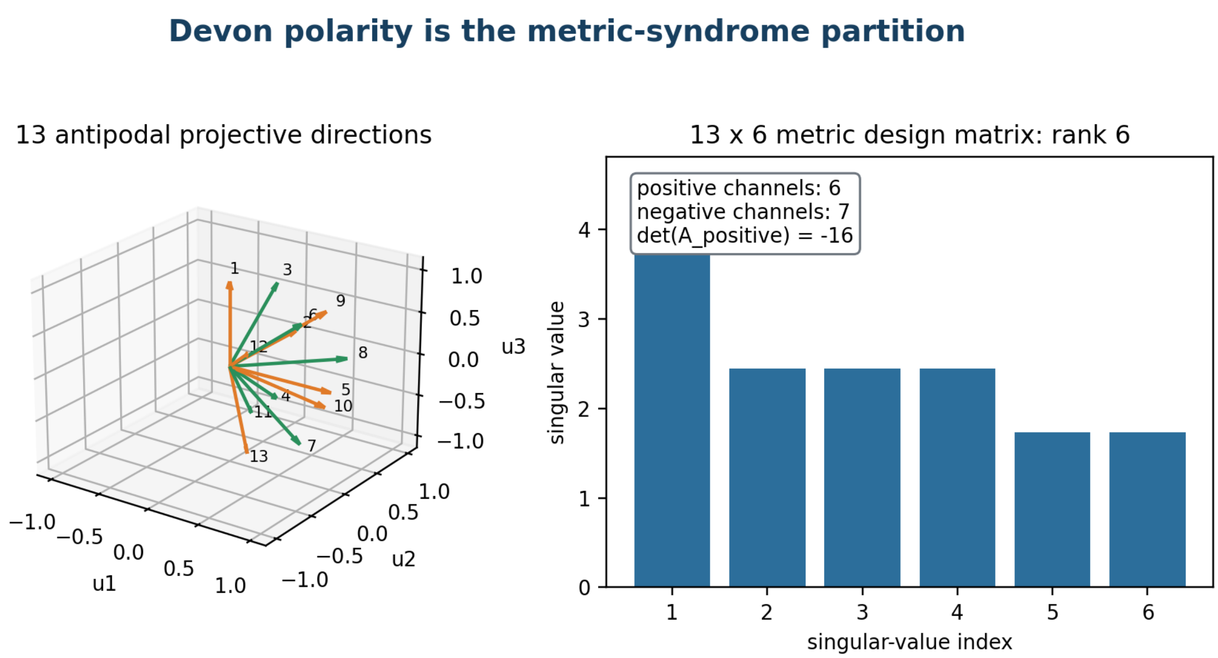

Figure 5.1. The left panel shows the complete 13-direction projective quotient. Green directions are the six positive face-diagonal metric channels; orange directions are the seven negative axis/body-diagonal syndrome channels. The right panel displays the six nonzero singular values of the complete 13 x 6 metric design matrix.

5.2 Six positive channels reconstruct the metric

A symmetric 3 x 3 metric has six independent entries. For each projective direction u, the measured quadratic channel is q(u)=u^T G u. The supplied 13 x 6 design matrix maps the six independent metric entries into the 13 projective quadratic observations. It has rank six. More strongly, the six rows selected by positive Devon polarity form an invertible 6 x 6 matrix with determinant -16.

```
rank(A)=6; rank(A_positive)=6; det(A_positive)=-16
```

Therefore the positive-polarity projective channels are a complete minimal metric decoder. The seven negative-polarity channels are not required to determine G; they are predicted after G is recovered and consequently provide seven independent consistency constraints. Devon polarity is thus exactly the organizing rule that separates metric information from error syndrome.

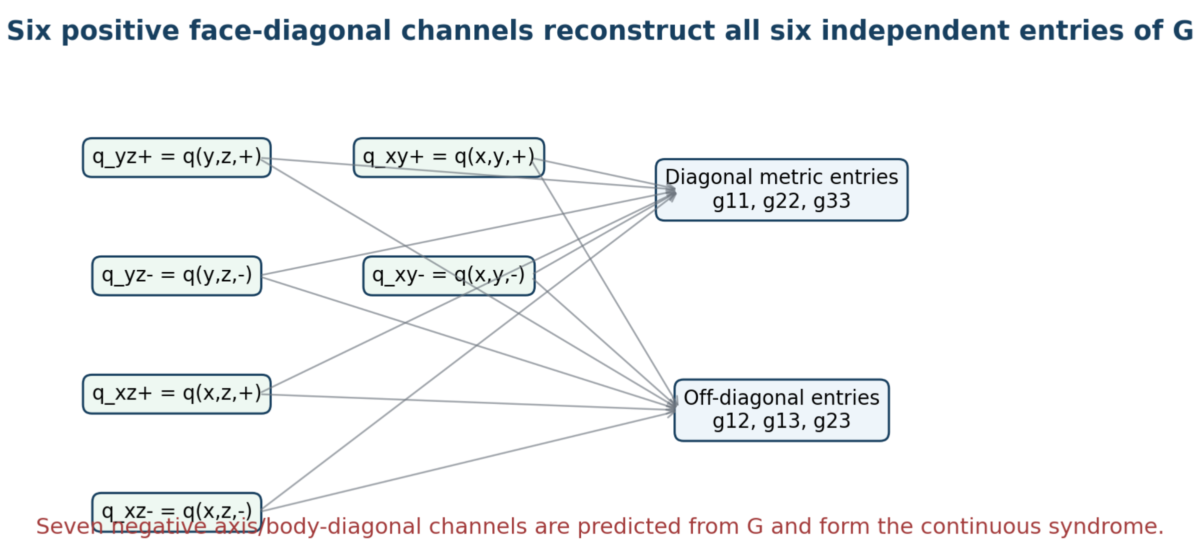

Figure 5.2. Information flow of the six-channel decoder. Sums of opposite face-diagonal measurements recover the three diagonal metric entries; differences recover the three off-diagonal entries. The seven remaining channels form the continuous syndrome.

6. Explicit Projective Metric Decoder

6.1 Channel definitions

Let q_yz^+ and q_yz^- denote q(0,1,1) and q(0,1,-1), with analogous definitions for the xz and xy pairs. Form the pair averages

```
S_yz=(q_yz^+ + q_yz^-)/2; S_xz=(q_xz^+ + q_xz^-)/2; S_xy=(q_xy^+ + q_xy^-)/2
```

The diagonal metric entries are

```
g11=(S_xz+S_xy-S_yz)/2
g22=(S_yz+S_xy-S_xz)/2
g33=(S_yz+S_xz-S_xy)/2
```

and the off-diagonal entries are

```
g23=(q_yz^+ - q_yz^-)/4
g13=(q_xz^+ - q_xz^-)/4
g12=(q_xy^+ - q_xy^-)/4
```

6.2 Syndrome prediction

After G is reconstructed, the three coordinate-axis observations must equal g11, g22, and g33. The four body-diagonal values must equal the corresponding signed sums of all diagonal and off-diagonal terms. Their differences from measured values define a seven-component real-valued syndrome vector.

```
syndrome = q_negative_measured - A_negative g_reconstructed
```

A zero syndrome certifies consistency of all 13 projective channels with one symmetric metric. A nonzero syndrome localizes measurement drift, channel corruption, or a physical departure from the six-parameter metric model. This is a continuous graph code: it performs both parameter reconstruction and error detection without discarding any projective direction.

7. Mass-Bridge as a Reciprocal Graph Invariant

7.1 Cofactors and reciprocal diagonal

The diagonal cofactors of the direct metric are the squared primitive face areas. Since R=G^-1, each reciprocal diagonal entry is the corresponding cofactor divided by det(G). Thus the registered Mass-Bridge face-area ratios can be read directly from reciprocal graph weights.

```
eta = 1000 sqrt(R11/R33)
alpha^-1 = 100 sqrt(R11/R22)
eta alpha = 10 sqrt(R22/R33)
```

Using the locked reciprocal metric gives eta=1836.152673426000, alpha^-1=137.035999177000, and eta alpha=13.399053419929. These reproduce the locked projective ratio system exactly within the recorded numerical precision.

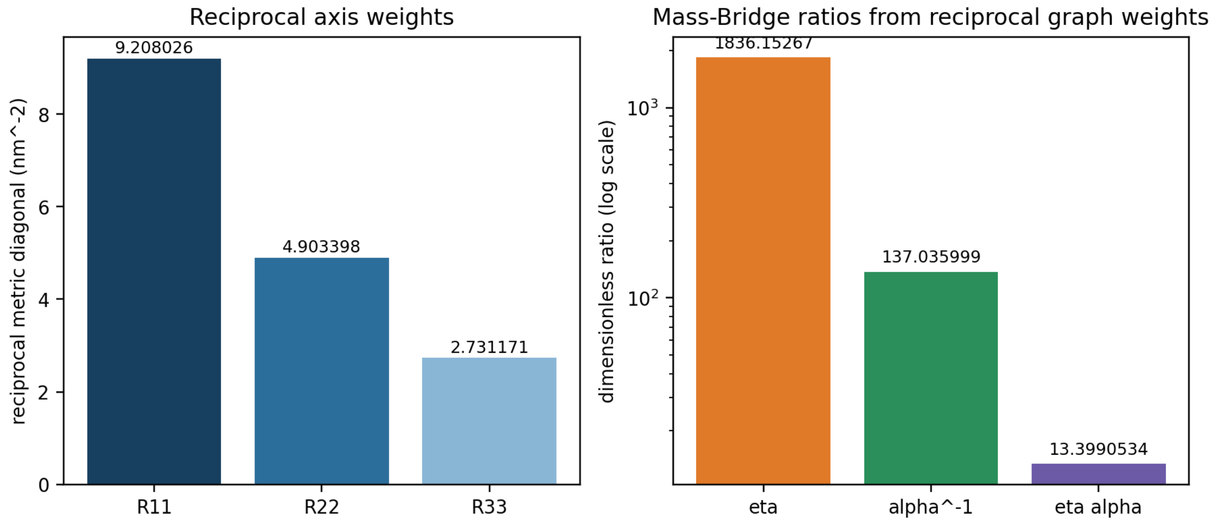

Figure 7.1. Reciprocal metric diagonal and the three Mass-Bridge ratios reconstructed from it. The ratios are dimensionless; the reciprocal weights have units nm^-2.

7.2 Weighted transport graph

Consider identical material coupling adjacent Cell3 volumes through their shared primitive faces. A conductance along channel a is proportional to face area A_bc divided by spacing h_a. Because h_a=V/A_bc, the channel weight is

```
c_a = sigma A_bc/h_a = sigma V R11
```

with cyclic expressions for b and c. Therefore the Mass-Bridge ratios are square-root ratios of directional graph conductances:

```
eta=1000 sqrt(c_a/c_c); alpha^-1=100 sqrt(c_a/c_b); eta alpha=10 sqrt(c_b/c_c)
```

This creates a direct experimental pathway. A resistor, acoustic, elastic, photonic, or oscillator network fabricated with Cell3 geometry should display directional spectral splitting determined by the locked reciprocal diagonal. Measuring that splitting provides an independent graph-domain reconstruction of the Mass-Bridge.

channel	face	spacing (nm)	Rii (nm^-2)	c_i/sigma = V Rii (nm)
a	bc	0.329546529695	9.20802552946	1.08576333503
b	ac	0.451597379721	4.90339831237	0.578183682003
c	ab	0.605097741519	2.73117127471	0.32204576565

8. Lucas Supergroup as the Exact 9B Graph Closure

8.1 Construction and census

A Lucas Group is a 3 x 3 x 3 block of Cell3 volumes. A Lucas Supergroup is the exact 9B oblique cubical complex containing 9 x 9 x 9=729 Cell3 volumes, equivalently 27 Lucas Groups. Its cellular census is

```
(f0,f1,f2,f3)=(1000,2700,2430,729)
```

and its additional graph invariants are 6,859 refined barycentric states, 55 laminar sheets, 1,944 internal interfaces, 1,946 boundary states, 486 exposed primitive faces, and Cell3-center graph cycle rank 1,216.

Object or invariant	count
Lucas Groups	27
Cell3 volumes	729
unrefined corner nodes	1000

Object or invariant	count
edges	2700
faces	2430
refined states	6859
laminar sheets	55
internal interfaces	1944
boundary states	1946
exposed primitive faces	486
center-graph cycle rank	1216

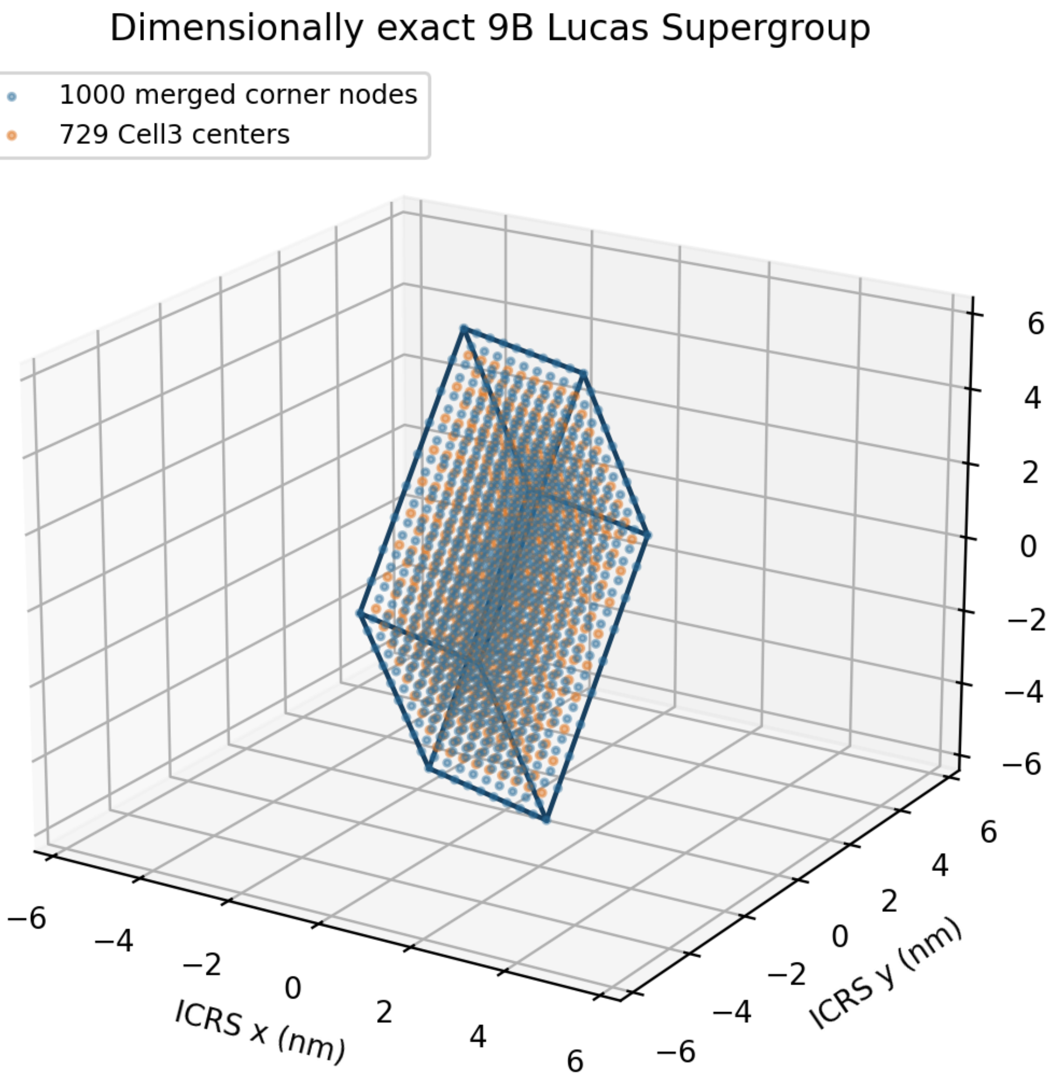

Figure 8.1. Dimensionally exact ICRS rendering of the 9B Lucas Supergroup. Blue points are the 1,000 merged unrefined corner nodes and orange points are the 729 Cell3 centers. The outer basis spans are generated directly from 9B.

8.2 Graph hierarchy inside the closure

Several graphs coexist inside the same 9B volume. The cellular chain complex has its own boundary operators and trivial homology because the solid block is contractible. The Cell3-center adjacency graph has 729 vertices and 1,944 internal-interface edges, giving cycle rank 1,944-729+1=1,216. The refined half-step lattice has 6,859 states and 55 laminar grades. These quantities should not be conflated; they answer different graph questions.

The exact identity 1,946=1,944+2 is the boundary-interface closure relation that uniquely selects side length nine in the isotropic family. At the same time, a 9 x 9 x 9 array of Lucas Supergroups reproduces the complete supergroup census at supergroup scale, forming exact second-order closure.

9. Devon Super Sheets and Emily Zero Modes

9.1 Exact sheet profile

At refined resolution a Lucas Supergroup has coordinates i,j,k from -9 through 9. A Devon Super Sheet is the set of states with constant lambda=i+j+k and common polarity chi=(-1)^lambda. The 55 sheets range from lambda=-27 to +27. Their populations are quadratic near the center and triangular near the extremes:

```
L_n(lambda)=3n^2+3n+1-lambda^2, for |lambda|<=n
L_n(lambda)=((3n-|lambda|+1)(3n-|lambda|+2))/2, for n<=|lambda|<=3n
```

For n=9 the central population is 271, the total is 6,859, and the signed sum is -1. At every exact closure order the positive and negative state counts differ by one, with net polarity -1; the boundary contributes -2 and the interior contributes +1.

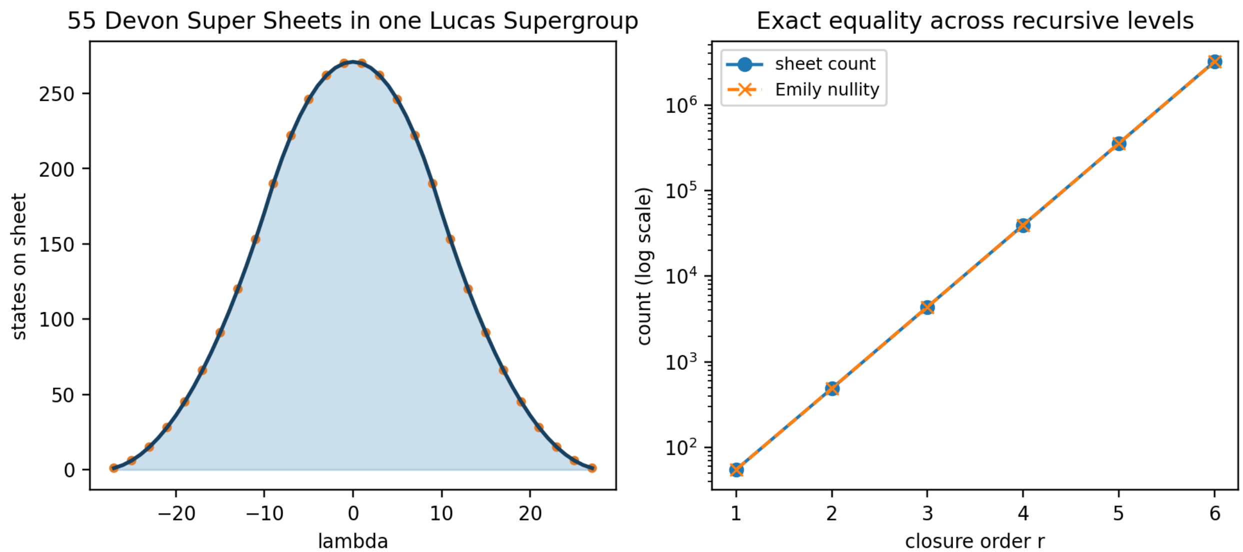

Figure 9.1. Left: the complete 55-sheet population profile for n=9, generated from the machine-readable sheet ledger. Right: equality of sheet count and Emily adjacency nullity across closure orders 1 through 6.

9.2 Emily nullity theorem

Let A_n be the nearest-neighbor adjacency operator of the open cubic graph P_n x P_n x P_n. Its eigenvalues are sums of three path-graph cosine eigenvalues. For the refined Lucas family n=18s+1, the zero modes consist of the central triplet and six permutations of complementary path indices. Their count is

```
nullity(A_n)=1+6(n-1)/2=3n-2
```

The number of laminar sheets in an odd n x n x n grid is also 3(n-1)+1=3n-2. Therefore the Emily zero-mode dimension equals the Devon Super Sheet count throughout the family.

```
dim ker(A_n) = number of Devon Super Sheets
```

At closure order r, the Cell3 side is 9^r and refined side is n=2*9^r+1, giving 6*9^r+1 sheets and the same number of zero modes. This equality identifies the Devon sheets as the cardinal structure of the zero-energy adjacency sector, not merely a visual partition.

9.3 Sheet-nullspace intertwiner

The equality of dimensions defines a new constructive problem: build an explicit linear map from sheet-amplitude space to the Emily nullspace. Such an intertwiner would associate one independent zero mode with each laminar grade while preserving reflection, polarity, and recursive inheritance. It would convert spatial sheet control into direct spectral mode control.

```
T_sheet_to_null : R^(6*9^r+1) -> ker(A_n)
```

10. Recursive Morphology as Dimension-Graded Renormalization

10.1 Four scaling channels

One exact Lucas closure step multiplies linear scale by nine. A d-dimensional geometric contribution therefore scales by 9^d. The volume, face, edge, and point channels are

```
729, 81, 9, 1
```

and the renormalization operator is diagonal in geometric dimension:

```
R_scale = diag(729,81,9,1)
```

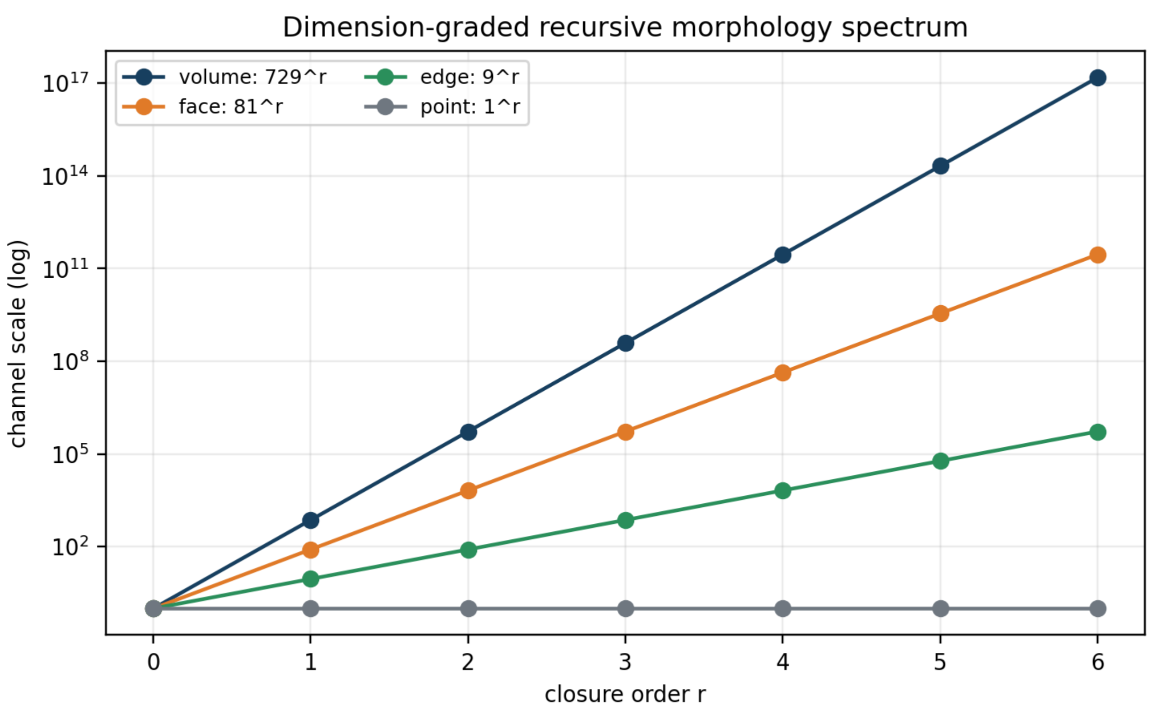

Figure 10.1. Recursive morphology channels through order six. The logarithmic vertical scale exposes the constant factors 729, 81, 9, and 1 for volume, face, edge, and point contributions.

10.2 Universal recurrence

The characteristic polynomial of the four scale factors is

```
(x-729)(x-81)(x-9)(x-1)=x^4-820x^3+67158x^2-597780x+531441
```

so every observable of the form X_r=A*729^r+B*81^r+C*9^r+D obeys

```
X_(r+4)=820 X_(r+3)-67158 X_(r+2)+597780 X_(r+1)-531441 X_r
```

The registered recursive energy occupies only the volume and face channels, E_r=3C*729^r-C*81^r, and therefore obeys the reduced second-order recurrence E_(r+2)=810E_(r+1)-59049E_r.

10.3 Dimensional tomography

Measurements collected across closure order can be decomposed into the four basis sequences. The coefficient of 729^r isolates bulk response; 81^r isolates surface response; 9^r isolates edge response; and the constant isolates point or topological response. A residual that cannot be represented by these four channels is evidence for an additional graph degree, coupling, or dynamical term. Recursive scaling therefore becomes a practical diagnostic instrument rather than a descriptive pattern.

11. Closure as an Exact Graph-Cut Balance

11.1 General rectangular closure

For an aligned a x b x c array of Lucas Supergroups, write S=ab+ac+bc. The coarse boundary-state and internal-interface counts are B_boundary=8S+2 and I_internal=3abc-S. Imposing B_boundary=I_internal+2 gives

```
1/a + 1/b + 1/c = 1/3
```

This is the complete rectangular closure equation. The machine-readable sweep contains 21 unordered positive integer solutions. The isotropic specialization a=b=c=s gives the unique solution s=9.

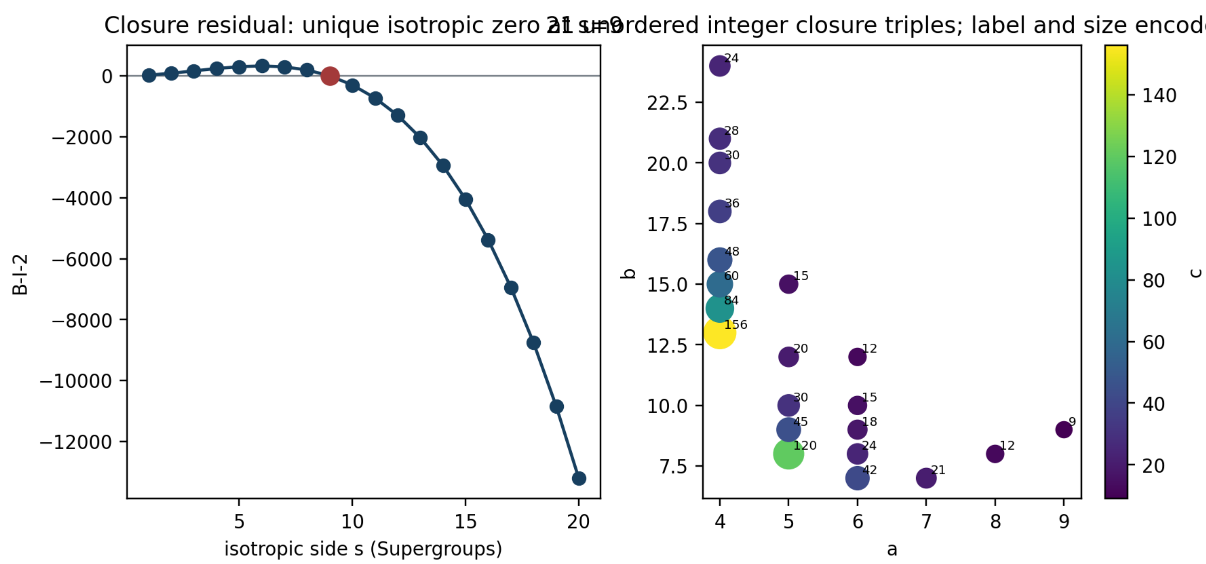

Figure 11.1. Left: the isotropic closure residual 3s^2(9-s), showing the unique zero at s=9. Right: all 21 unordered integer closure triples; each plotted label is c and marker size also increases with c.

11.2 Closure residual as graph flux

For isotropic side s, the exact residual is

```
Delta_closure = B_boundary - I_internal - 2 = 3s^2(9-s)
```

It is positive below s=9, zero at closure, and negative above closure. The sign distinguishes boundary-dominated from interface-dominated morphology. This residual acts as a discrete graph flux across scale and supplies a natural control variable for closure experiments.

a	b	c	N=abc	check 1/a+1/b+1/c
4	13	156	8112	0.333333333333
4	14	84	4704	0.333333333333
4	15	60	3600	0.333333333333
4	16	48	3072	0.333333333333
4	18	36	2592	0.333333333333
4	20	30	2400	0.333333333333
4	21	28	2352	0.333333333333
4	24	24	2304	0.333333333333
5	8	120	4800	0.333333333333
5	9	45	2025	0.333333333333
5	10	30	1500	0.333333333333
5	12	20	1200	0.333333333333
5	15	15	1125	0.333333333333
6	7	42	1764	0.333333333333
6	8	24	1152	0.333333333333
6	9	18	972	0.333333333333
6	10	15	900	0.333333333333
6	12	12	864	0.333333333333
7	7	21	1029	0.333333333333
8	8	12	768	0.333333333333
9	9	9	729	0.333333333333

12. Scalar Serialization Preserves Identity but Not Locality

12.1 Signed scalar address

The serialized address coordinate assigns a signed quadratic radius to each lattice address:

```
rho(k)=sigma(k) sqrt(k^T G k)
```

Under the injectivity condition that equal quadratic values occur only for antipodal pairs, rho is a directed bijection. It stores a hidden rank-three relational address in one explicit scalar coordinate. Addition in scalar space is therefore nonlinear and is defined by decoding to the lattice, adding addresses, and encoding again.

12.2 Measured nonlocality of the 27-node comb

Identity preservation does not imply geometric locality. Sorting the actual 27 local nodes by rho and comparing that order with the 54 physical nearest-neighbor edges gives only eight edges joining consecutive scalar ranks. The mean rank jump is 5.8148148, the median is 4, and the maximum is 23. By primitive direction the mean jumps are 7.3333 along a, 4.1111 along b, and 6.0000 along c.

Thus rho is an address coordinate rather than a distance coordinate. The scalar-order graph and physical-adjacency graph should be superposed. A physical edge with a large scalar-rank separation is a serialized-space shortcut carrying substantial relational information in one geometric step.

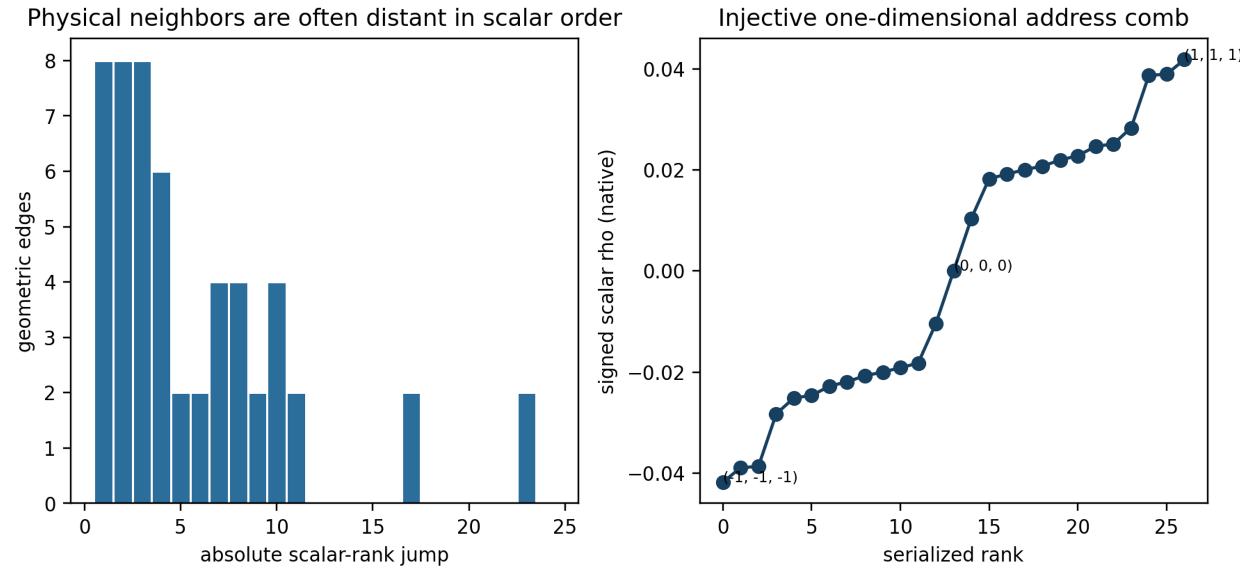

Figure 12.1. Left: exact distribution of scalar-rank jumps over all 54 physical Cell3 edges. Right: the complete signed scalar comb in serialized order. The data are taken from the supplied 27-node registry.

13. Integrated Discovery Pathways

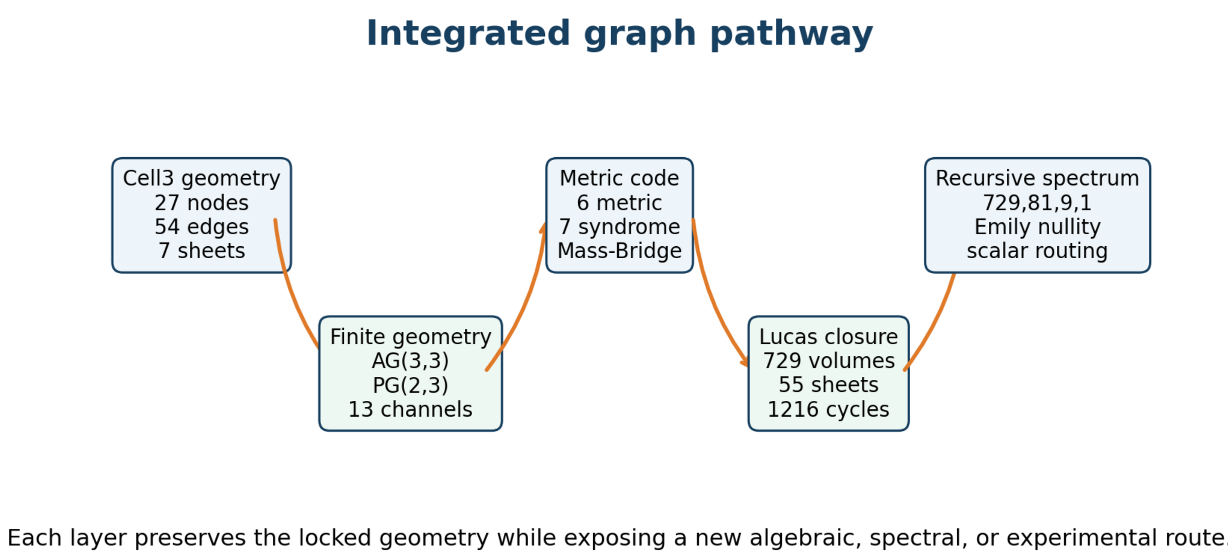

Figure 13.1. The integrated graph pathway from locked Cell3 geometry through finite incidence, metric decoding, Lucas closure, and recursive spectral routing.

13.1 Projective metric decoder

Measure the six positive face-diagonal quadratic channels, reconstruct all six entries of G with the explicit formulas, and predict the seven negative channels. The residual syndrome provides continuous error detection. Repeating the procedure over time turns the projective graph into a metric drift sensor.

13.2 Mass-Bridge spectral network

Fabricate a network whose directional couplings are proportional to V R11, V R22, and V R33. Electrical, acoustic, elastic, and coupled-oscillator realizations are all compatible with the graph law. The directional eigenfrequency or impedance splitting should reproduce the three Mass-Bridge ratios through square-root coupling ratios.

13.3 Sheet-nullspace control

Construct the explicit sheet-to-nullspace intertwiner. Drive or read out the 6*9^r+1 laminar sheets and determine whether the corresponding Emily zero modes can be addressed independently. Recursive closure predicts exact growth of the controllable zero-mode family.

13.4 Affine-projective flag graph

Extend the 27 affine points into the complete AG(3,3) incidence system. It contains 117 affine lines and 39 affine planes. There are 468 line-plane incidences and 1,404 point-line-plane flags. These flags create a richer graph for routing, coding, and symmetry analysis. The projective quotient then supplies direction classes for every affine line family.

13.5 Recursive dimensional tomography

Measure the same observable at multiple closure orders and fit the exact basis 729^r, 81^r, 9^r, and 1. The coefficients isolate bulk, surface, edge, and point contributions. Structured residuals identify new graph channels that are absent from the four-dimensional morphology model.

13.6 Scalar shortcut mining

Rank geometric edges and paths by scalar-order separation, projective class, sheet crossing, and reciprocal cost. The highest-separation edges are candidates for efficient relational routing, error-correcting transitions, and path-history-sensitive experiments.

13.7 Automorphism-orbit splitting

Begin with the full unweighted symmetry of PG(2,3), then apply the locked direct or reciprocal metric as a weight. Determine how the 13 channels split into weighted orbits. The resulting orbit hierarchy will identify which symmetries survive the physical metric and which are broken into distinct response classes.

14. Experimental and Computational Program

Program	Input graph	Primary observable	Exact prediction
six-channel metric reconstruction	six positive projective channels	reconstructed G and seven-component syndrome	det(A_positive)=-16; zero syndrome for one consistent metric
weighted transport network	Cell3 adjacency with c_i proportional to V R_ii	directional impedance or resonance splitting	Mass-Bridge ratios from square-root coupling ratios
sheet-mode experiment	refined P_n^3 adjacency	zero-frequency or zero-eigenvalue mode count	nullity=6*9^r+1
recursive tomography	same observable at successive closure orders	four scale coefficients and residual	basis sequences 729^r,81^r,9^r,1
closure sweep	rectangular a x b x c arrays	boundary-interface residual	closure iff 1/a+1/b+1/c=1/3
serialized shortcut analysis	physical edges plus scalar rank	rank-jump and path information	8 consecutive edges; mean jump 5.8148148; maximum 23

14.1 Data discipline

Each experiment should preserve the distinction among address, physical coordinate, graph adjacency, projective incidence, polarity, and scalar order. The same pair of states may be adjacent in one graph and remote in another. Measurements should therefore be stored with the full multiplex address: ternary coordinates, physical coordinates, lambda, chi, projective representative, direct q, reciprocal q, closure order, and scalar rho.

14.2 Falsifiable graph signatures

The strongest signatures are discrete and overdetermined: six positive channels must reconstruct one metric; seven negative channels must agree with that metric; transport weights must reproduce reciprocal ratios; sheet count must match Emily nullity; closure must occur only on the integer reciprocal-sum surface; and recursive observables must satisfy the fixed recurrence. These provide exact targets rather than proximity-only comparisons.

15. Central Result and Paradigm-Level Synthesis

The combined corpus reduces to one coherent graph hierarchy:

```
AG(3,3) -> PG(2,3) -> six metric channels + seven syndrome channels
-> 729-volume Lucas closure -> Devon-sheet/Emily-nullspace equivalence
-> {729,81,9,1} recursive spectrum -> scalar nonlocal routing
```

The strongest discovered relationship is that Devon polarity is the exact organizing principle of the 13-channel metric code. The six positive projective channels carry the complete symmetric metric. The seven negative projective channels carry the complete overdetermination syndrome. The Mass-Bridge constants are reciprocal invariants reconstructed from that metric. The Lucas hierarchy propagates the same organization through laminar sheets, zero modes, interfaces, closure balance, and recursive dimensional scaling.

This architecture changes the working scientific model from a collection of separate geometric, spectral, and ratio observations into a single relational graph system. Local geometry, finite projective incidence, physical metric, error correction, closure, and recursion become mutually constraining views of one object. The new pathways arise precisely because information that is redundant in one view becomes operational in another: polarity becomes a decoder partition, reciprocal geometry becomes a transport law, sheet count becomes nullity, closure becomes graph flux, and scalar nonlocality becomes a routing resource.

Appendix A. Complete 27-Node Cell3 Ledger

address	lambda	chi	class	x (nm)	y (nm)	z (nm)
(-1,-1,-1)	-3	-1	corner	-0.176351350136	0.118682429905	0.637311498325
(-1,-1,0)	-2	1	edge center	-0.181416008950	-0.076211421278	0.292444294142
(-1,0,-1)	-2	1	edge center	-0.158685962215	0.222342355125	0.363610955631
(0,-1,-1)	-2	1	edge center	-0.012600729109	0.091233925963	0.618567746878
(-1,-1,1)	-1	-1	corner	-0.186480667763	-0.271105272461	-0.052422910042
(-1,0,0)	-1	-1	face center	-0.163750621028	0.027448503942	0.018743751447
(-1,1,-1)	-1	-1	corner	-0.141020574293	0.326002280345	0.089910412937
(0,-1,0)	-1	-1	face center	-0.017665387922	-0.103659925220	0.273700542694
(0,0,-1)	-1	-1	face center	0.005064658813	0.194893851183	0.344867204184
(1,-1,-1)	-1	-1	corner	0.151149891919	0.063785422021	0.599823995431
(-1,0,1)	0	1	edge center	-0.168815279841	-0.167445347241	-0.326123452736
(-1,1,0)	0	1	edge center	-0.146085233106	0.131108429162	-0.254956791247
(0,-1,1)	0	1	edge center	-0.022730046735	-0.298553776403	-0.071166661489
(0,0,0)	0	1	body center	0.000000000000	0.000000000000	0.000000000000
(0,1,-1)	0	1	edge center	0.022730046735	0.298553776403	0.071166661489
(1,-1,0)	0	1	edge center	0.146085233106	-0.131108429162	0.254956791247
(1,0,-1)	0	1	edge center	0.168815279841	0.167445347241	0.326123452736
(-1,1,1)	1	-1	corner	-0.151149891919	-0.063785422021	-0.599823995431
(0,0,1)	1	-1	face center	-0.005064658813	-0.194893851183	-0.344867204184
(0,1,0)	1	-1	face center	0.017665387922	0.103659925220	-0.273700542694
(1,-1,1)	1	-1	corner	0.141020574293	-0.326002280345	-0.089910412937
(1,0,0)	1	-1	face center	0.163750621028	-0.027448503942	-0.018743751447
(1,1,-1)	1	-1	corner	0.186480667763	0.271105272461	0.052422910042
(0,1,1)	2	1	edge center	0.012600729109	-0.091233925963	-0.618567746878
(1,0,1)	2	1	edge center	0.158685962215	-0.222342355125	-0.363610955631
(1,1,0)	2	1	edge center	0.181416008950	0.076211421278	-0.292444294142
(1,1,1)	3	-1	corner	0.176351350136	-0.118682429905	-0.637311498325

Appendix B. Complete Six-Plane Ledger

channel	face	sign	normal ICRS	area nm^2	spacing nm	nine node addresses
a	bc	-1	(-0.995965190309,0.083602162507,-0.032619290564)	0.357809539129	0.329546529695	(-1,-1,-1) (-1,-1,0) (-1,-1,1) (-1,0,-1) (-1,0,0) (-1,0,1) (-1,1,-1) (-1,1,0) (-1,1,1)
a	bc	1	(-0.995965190309,0.083602162507,-0.032619290564)	0.357809539129	0.329546529695	(1,-1,-1) (1,-1,0) (1,-1,1) (1,0,-1) (1,0,0) (1,0,1) (1,1,-1) (1,1,0) (1,1,1)
b	ac	-1	(-0.089052594102,-0.866576769291,0.491033948326)	0.261106235790	0.451597379721	(-1,-1,-1) (-1,-1,0) (-1,-1,1) (0,-1,-1) (0,-1,0) (0,-1,1) (1,-1,-1) (1,-1,0) (1,-1,1)
b	ac	1	(-0.089052594102,-0.866576769291,0.491033948326)	0.261106235790	0.451597379721	(-1,1,-1) (-1,1,0) (-1,1,1) (0,1,-1) (0,1,0) (0,1,1) (1,1,-1) (1,1,0) (1,1,1)
c	ab	-1	(0.194092200808,0.913177168821,0.358379234791)	0.194869165460	0.605097741519	(-1,-1,-1) (-1,0,-1) (-1,1,-1) (0,-1,-1) (0,0,-1) (0,1,-1) (1,-1,-1) (1,0,-1) (1,1,-1)
c	ab	1	(0.194092200808,0.913177168821,0.358379234791)	0.194869165460	0.605097741519	(-1,-1,1) (-1,0,1) (-1,1,1) (0,-1,1) (0,0,1) (0,1,1) (1,-1,1) (1,0,1) (1,1,1)

Appendix C. Complete 13-Channel Projective Ledger

id	representative	chi	role	q_direct nm^2	q_recip nm^-2	incidences
1	(0,0,1)	-1	syndrome	0.627770610077	2.731171274712	2,5,8,11
2	(0,1,0)	-1	syndrome	0.343877732393	4.903398312368	1,5,6,7
3	(0,1,1)	1	metric	1.564433860393	3.004218418247	4,5,10,12
4	(0,1,-1)	1	metric	0.378862824547	12.264920755912	3,5,9,13
5	(1,0,0)	-1	syndrome	0.111676057896	9.208025529458	1,2,3,4
6	(1,0,1)	1	metric	0.827321138166	10.648820732769	2,7,10,13
7	(1,0,-1)	1	metric	0.651572197781	13.229572875569	2,6,9,12
8	(1,1,0)	1	metric	0.496974456850	14.114493446682	1,11,12,13
9	(1,1,1)	-1	syndrome	1.805405055043	10.924937481161	4,7,9,11
10	(1,1,-1)	-1	syndrome	0.444085078812	22.766391961626	3,6,10,11
11	(1,-1,0)	1	metric	0.414133123728	14.108354236969	1,8,9,10
12	(1,-1,1)	-1	syndrome	0.536992686074	20.179500609114	3,7,8,12
13	(1,-1,-1)	-1	syndrome	1.546814781536	13.499550414248	4,6,8,13

Appendix D. Recursive Closure Ledger

order	Cell3/axis	Cell3 volumes	Lucas Supergroups	Q refined	sheets/nullity	interfaces	boundary states	cycle rank
1	9	729	1	6859	55	1944	1946	1216
2	81	531441	729	4330747	487	1574640	157466	1043200
3	729	387420489	531441	3105745579	4375	1160667144	12754586	773246656
4	6561	282429536481	387420489	2259952891867	39367	847159469280	1033121306	564729932800
5	59049	205891132094649	282429536481	1647170898524299	354295	617662935930744	83682825626	411771803836096
6	531441	150094635296999121	205891132094649	1200760471533619387	3188647	450283058602387920	6778308875546	300188423305388800

Appendix E. Formula Ledger

Concept	Formula
Cell3 node	r(u)=B u/2; u in {-1,0,1}^3
laminar grade and polarity	lambda=u1+u2+u3; chi=(-1)^lambda
open graph	P3 x P3 x P3; V=27; E=54; beta1=28
projective quotient	(F3^3-{0})/{+/-1}=PG(2,3); 13 classes
incidence identity	N^2=3I+J
projective transform	M=3N-J; M^2=27I-2J
quadratic channel	q(u)=u^T G u
signed scalar	rho(k)=sigma(k)sqrt(k^T G k)
Mass-Bridge reciprocal	eta=1000sqrt(R11/R33); alpha^-1=100sqrt(R11/R22); eta alpha=10sqrt(R22/R33)
transport weights	c_a=sigma V R11; c_b=sigma V R22; c_c=sigma V R33
Lucas census	(f0,f1,f2,f3)=(1000,2700,2430,729)
sheet/nullity	S_r=dim ker A_n=6*9^r+1
recursive observable	X_r=A729^r+B81^r+C9^r+D
universal recurrence	X_(r+4)=820X_(r+3)-67158X_(r+2)+597780X_(r+1)-531441X_r
closure equation	1/a+1/b+1/c=1/3
isotropic residual	Delta=3s^2(9-s)

Appendix F. Source and Reproducibility Record

All numerical figures, tables, and derived identities in this report were generated from the machine-readable source files listed in Document Control. The locked physical basis and planes come from the Mass-Bridge projective ratio system. Projective representatives, incidence, transform, design matrix, and scalar comb come from the serialized spectral address system. Recursive closure levels and morphology channels come from the recursive Lucas spectrum. Sheet populations and polarity laws come from the Devon Super Sheet record. Rectangular closure solutions and large-scale laws come from the Lucas Supergroup morphology record.

The companion JSON records the exact source hashes, numerical ledgers, derived graph invariants, equations, and figure hashes required to reproduce this report.